DR. A. P. J. ABDUL KALAM TECHNICAL UNIVERSITY
LUCKNOW, UTTAR PRADESH



SYLLABUS & EVALUATION SCHEME WITH SYLLABUS

FOR

B. TECH. 4th YEAR

PLASTIC ENGINEERING

AS PER

AICTE MODEL CURRICULUM
[Effective from the Session: 2021-22]

B. Tech Plastic Engineering
Evaluation Scheme
Effective in Session 2021-22

SEMESTER- VII

S. No.	Code	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1		HSMC-1/HSMC-2	3	0	0	30	20	50		100		150	3
2	KPE 071 - 074	Departmental Elective-IV	3	0	0	30	20	50		100		150	3
3	KPE 075 - 078	Departmental Elective-V	3	0	0	30	20	50		100		150	3
4		Open Elective-II	3	0	0	30	20	50		100		150	3
5	KPE 751	**Plastic Recycling & Waste Management**	0	0	2				25		25	50	1
6	KPE 752	Mini Project or Internship Assessment*	0	0	2				50			50	1
7	KPE 753	Project	0	0	8				150			150	4
8		MOOCs (Essential for Hons. Degree)											
		Total	**9**	**0**	**12**	**21**						**850**	**18**

*The Mini Project or internship (5 - 6 weeks) conducted during summer break after VI semester and will be assessed during VII semester.

SEMESTER- VIII

S. No.	Code	Subject	Periods			Evaluation Scheme				End Semester		Total	Credit
			L	T	P	CT	TA	Total	PS	TE	PE		
1		HSMC-2/HSMC-1	3	0	0	30	20	50		100		150	3
2		Open Elective-III	3	0	0	30	20	50		100		150	3
3		Open Elective-IV	3	0	0	30	20	50		100		150	3
4	KPE 851	Project	0	0	18				100		300	400	9
5		MOOCs (Essential for Hons. Degree)											
		Total	**9**	**0**	**18**	**27**						**850**	**18**

Students can choose any elective horizontally from the pool of electives.

	Specialization -1	Specialization -2	Specialization -3	Specialization -4
Sem VII Code	**KPE 071**	**KPE 072**	**KPE 073**	**KPE 074**
Departmental Elective-IV	Polymer composites	Biomedical Plastics	Conducting Polymers	Polymer Structure Property relationship
Sem VII Code	**KPE 075**	**KPE 076**	**KPE 077**	**KPE 078**
Departmental Elective-V	Biodegradable Polymers	Rubber Technology	Polymer Nanomaterials	Thermoplastic Elastomers

Subject Code: KPE 751	Plastic Recycling & Waste Management	L T P : 0 0 2	Credits: 1

Course outcomes: The students will be able to		Blooms Taxonomy
CO1	Environmental impact due to misuse of plastic product	K2
CO2	Practical knowledge of plastic recycling & waste Management	K1
CO3	Hands-on experience of Various process involve for plastic recycling.	K3
CO4	Analyse the effect recycling by chemical modification of plastic waste	K3
CO5	Determine the optimum recycling by co-extrusion and co-injection modeling	K2

Study of the following (At least 8 of the following)

S.No.	Name of the Experiment
1	To study various sources of plastic waste
2	To study waste disposal method.
3	To perform segregation of plastic waste by simple identification methods
4	To perform plastic waste size reduction by mechanical method
5	To carry out separation of plastics using float-sink method
6	To carry out agglomeration of plastic film waste
7	To perform primary recycling of plastic film waste
8	To study recycling by chemical modification of plastic waste
9	To study secondary recycling by co-extrusion and co-injection modling

Semester – VII: Departmental Elective – IV

Subject Code: KPE 071	Polymer Composites	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Understand the concepts of polymer composites and their applications	K2
CO-2	Understand of theories of composite formation and mechanism of load transfer	K2
CO-3	Acquire knowledge of various techniques for compounding of thermoplastics.	K2
CO-4	Understand the process of manufacture of fibre reinforced plastics	K2
CO-5	Acquire knowledge of testing & characterisation of composites	K2

UNIT I

Introduction to composite materials

Classification - advantages - polymer composites. Thermosetting and thermoplastic matrix materials – Reinforcements-Types and Forms-particulate, flake, fibrous etc.- coupling agents-Principles of composite reinforcement. Effect of reinforcement on composite properties.

UNIT II

Theory of composite materials

Calculation of composite properties- mechanism of load transfer,
minimum and critical fibre content, critical fibre length- Rule of mixtures – Halpin -Tsai - equation.

UNIT III

Compounding of Thermoplastics

Twin screw extrusion, compression moulding- compounding of
polyolefins, polystyrene and styrene copolymers, engineering polymers, wood floor and natural fiber filled plastics, compounding lines, post compounding operations.

UNIT IV

FRP processing

Important methods - hand layup, spray up, filament winding, compression moulding, injection moulding, resin, transfer moulding, reaction injection moulding, pultrusion,

Miscellaneous Methods

Machinery, operation, advantages and disadvantages.

UNIT V

Characterization of Composites

Control of particle/fibre and porosity content, particle/fibre distribution, Interfacial Reaction of matrix-reinforcing component, Coating of reinforcing component, Strength analysis

Testing Quality control & end use of Composites

Testing for mechanical, electrical, thermal, optical and chemical properties, Determination of shelf life and gel time – Non-destructive testing methods.

Application of composite
Marine, chemical, railways, electrical and electronic industry, space structures, Robotics.

Text Books:-

1. J.A.Brydson, "Plastics materials", Butterworth- Heinemann - Oxford, 6th Ed., 1995.
2. S.T.Peters, “Handbook of Composites”, Chapman & hall, 2nd Edition 1998.

References:-

1. G Lubin, “Hand Book of Composites”, 2nd Ed, Van Nostrand Reinhold, New York, 1982.
2. F.L. Matthews and R.D. Rawlings, 'Composite materials: engineering and science', Chapman and Hall, 1994.
3. P.K. Mallick, 'Composites Engineering Handbook', Marcel Dekker Inc.NY., 1997.
4. D. Hull and T. W. Clyne, "An introduction to Composite Materials 2nd Ed", Cambridge, 1996
5. Astrom; B.T, Manufacture of Polymer Composites, Chapman and Hall, London(1997).

Semester – VII: Departmental Elective – IV

Subject Code: KPE 072	Biomedical Plastics	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Acquire knowledge of nature and application of biomedical plastics	K2
CO-2	Understand various aspects of natural biomedical polymers	K2
CO-3	Acquire knowledge about various applications of polymers in biomedical devices	K2
CO-4	Understand the various aspect of application of polymers in soft lenses	K2
CO-5	Understand the various aspect of biopolymers in dental field	K2

UNIT I

Introduction to biomedical applications of polymers;

Contemporary biomedical materials, Their advantages over other materials in use.

Biomaterials

Biomaterials, Bio-Compatibility, Stabilization, Inflammation and Wound Healing, Blood Clotting System, Kinn System, Biological response to Implants, Implant Design and Applications.

UNIT II

Biomedical Polymers

Natural biomedical polymers (natural rubber, representative biopolymers) Biodegradation and biodegradable polymers like polylactides, polyglycolides and their copolymers; polydioxanones etc., Criteria for the Selection of Biomedical Polymers, Physicochemical Aspects of the Blood Compatibility of Polymeric Surface. Biomedical Polymers from biological source, Poly hydroxy Alkanoic Acids, Microbial polysaccharides, Silk, Collagen, Microbial Cellulose, Hyaluronic Acid, Synthetic Polymers such as PTMA, Silicon Rubber, Polyethylene, Natural Rubber, Hydrogels.

UNIT III

Biomedical Applications of Polymers

Permanent implants For Function- Orthopaedics, Cardio Vascular, Respiratory Patches and Tubes, Digestive System, Genitourinary System, Nervous System, Orbital (Corneal And Lens Prosthesis) – Permanent Implant For Cosmoses, Other Applications of Engineered Material In Clinical Practices, Silicone Implants, Polymer Membranes, drug delivery systems, Polymer Skin, Polymeric Blood substitute. Properties and applications of various polymers used for biomedical polymers Synthetic biomedical polymers [Polyolefins; PVC, Polyacrylates, Polystyrenes and its copolymers, Polyesters, Polyamides (Nylons), Fluorocarbon polymers, Polyacetals, Polycarbonates, Polysulphones, Epoxy resins]

UNIT IV

Polymeric Lenses

Contact Lenses, Hard Lenses, Gas Permeable Lenses, Flexible Lenses, Soft Lenses, Hydrogels, Equilibrium Swelling, Absorption and Desorption, Oxygen Permeability, Types of Soft Lenses, Manufacture, Cleaning And Disinfection.

UNIT V

Dental Polymers

Dental applications, denture bases, dentate reliners, crown and bridge resins, plastic teeth, mouth protectors, maxillofacial prosthetic materials, restorative material, polyelectrolyte based restoratives, sealants, adhesives, dental impression and duplicating materials, agar, algometer elastomers.

Reference Books:

1. H.F. Mark (Ed), Encyclopedia of polymer science and engineering, John Wiley and Sons New York, 1989.
2. Galin and M. Ruben Ed., Soft compact Lenses clinical and applied Technology. John Wiley and Sons, Inc. New York, 1978.
3. HR Allcock, FW Lampe and JE Mark. Contemporary polymer chemistry (3rd edition). Pearson Education, Inc (Pearson/Prentice Hall)
4. Manas Chanda, Salil K. Roy (Ed) Plastic Technology Hand Book Marcel Dekker, Inc. New York, 1993.

Semester – VII: Departmental Elective – IV

Subject Code: KPE 073	Conducting Polymers	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Understanding of the basics of conducting polymers and their conduction mechanism	K2
CO-2	understand various types of conducting polymers and their properties	K2
CO-3	Aquire knowledge about various mechanisms and techniques used in synthesis of conducting polymers	K2
CO-4	Understand various characterisation techniques for conducting polymers	K2
CO-5	Acquire knowledge of various applications of conducting polymers	K2

UNIT I

Introduction

Need of conducting polymers, Classification of conducting polymer, Concept of doping,n-Type, p-Type, Electrochemistry of electronically conducting polymers-source of electronic conduction in polymers, polaron, bipolaron, conduction mechanism.

UNIT II

Properties of conducting polymers

Structure-property relationship, Types of conducting polymers, e.g. Polyaniline (PANI), Polypyrrole (PPy), Polythiophene (PTh), Discovery of polyacetylene

UNIT III

Synthesis of conducting polymer-Chemical synthesis, electrochemical synthesis, template synthesis, precursor synthesis, soluble polymers (Colloid and dispersion), advantage and disadvantage of various synthesis methods. General Methodology; Synthesis and processability of selected conducting polymers like – Polyacetylene, Polyaniline, Polypyrrole, Polythiophene and Poly-para – phenylene

UNIT IV

Analytical Techniques for Characterization of Conducting polmers

IR ,UV, Impedance spectroscopy, Fourier Transform Infra-red spectroscopy, X-ray photoelectron spectroscopy**,** Scanning Electron microscopy **(**SEM)**,** Transmission electron microscopy(TEM), Electrochemical quartz crystal micro balance(EQCM), Four Probe conductivity measurement, Galvanostat/Potentiostat

UNIT V

Applications

Rechargeable batteries, o-LED, Gas sensors, Bio sensors, Photovoltaic energy device, Microelectronics, PCB fabrication, Electro catalyst. Application proposed antistatic coating, electrochemical mechanical device, super capacitor, Telecommunication system, Electromagnetic screening material, Analytical sensor.

Recent trend in conducting polymer, functionalized conducting polymer (Second generation polymer), Super conductor (Inorganic, organic hybrid structure), Conducting polymer based on nano composite.

Text book :-

1. Electrochemical science and technology of polymers -1 & 2 ed., R.G Linford, Elsevier applied sciences, London 1987 and 1989.

Reference Books:

1. Hand book of Conducting Polymers: Terje A. Skotheim (Vol.l), Dekker (668.42)
2. Hand book of Polymer Synthesis (Part B): Hans Kricheldorf, Dekker (668.9).
3. Sensors: Principles and Applications: Peter Hauptmann, Prentice Hall
4. Polymer Science and Technology: Premamoy Ghosh, Tata McGraw Hill (668.42).
5. Hand book of organic conductive molecules and polymers, Hari singh Nalwa (ed.), 4-Volume set, John wiley and sons, England 1997.

Semester – VII: Departmental Elective – IV

Subject Code: KPE 074	Polymer Structure And Property Relationship	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Understand the basics of structure and properties of polymers.	K2
CO-2	Acquire knowledge about effect of polymer structure on various properties of polymers.	K2
CO-3	Acquire knowledge of different thermal transitions in polymers	K2
CO-4	Understand various electrical and optical properties of polymers	K2
CO-5	Understand various chemical properties of polymers	K2

UNIT I

Structure of polymers

Linear, branched, cross-linked and network polymers – Homochain and hetero atomic chain polymers-Co-polymers-Linear and cyclic arrangement – Prediction of polymer properties, group contribution techniques, topological techniques- Volumetric properties-molar volume, Density, Vander Waals volume - Coefficient of linear thermal expansion and volumetric thermal expansion – Pressure volume temperature (PVT) relationship.

UNIT II

Stress-strain properties of polymers

Effect of polymer structure on modulus of elasticity, tensile strength, flexural strength, impact strength, yield strength, fracture toughness –Crazing in glassy polymers - Ductile brittle transition. Effect of additives on mechanical properties of polymers - Creep, stress relaxation and fatigue.

UNIT III

Thermodynamic and transition properties

Transition temperature in polymers, glass transition (Tg), melt transition (Tm), relationship between Tg and Tm – other transitions like ß-transitions, upper and lower glass transition temperatures-Prediction of Tg and Tm of polymers by group contributions.

Calorimetric properties – Heat capacity, specific heat, latent heat of crystallization and fusion, enthalpy and entropy – Calculation of heat capacities of polymers.

UNIT IV

Electrical and optical properties

Effect of polymer structure on Dielectric constant, power factor, dissipation factor and loss factor effect of frequency of voltage and temperature on dielectric properties - Prediction of molar polarization and effective dipole moment. Effect of additives on electrical properties of polymers. Optical properties-Effect of polymer structure on optical properties-clarity, transparency, haze, transmittance, reflectance and gloss – Prediction of refractive indices of polymers by group contributions.

UNIT V

Chemical Properties

Cohesive energy, cohesive energy density, solubility parameter, determination of solubility parameter of polymers – Prediction of solubility parameter- Effect of polymer structure on solubility in solvents and oils - Influence of structure in prediction of flame retardancy, water repellency – Chemical resistance of polymers - Polymer toxicity.

Reference :-

1. D.W. Van Krevelen And P.J. Hoftyzen, "Properties Of Polymer , 3rd Edition Elsevier Scientific, Publishing Company Amsterdam - Oxford - Newyork . 1990.
2. J.E. Mark Ed.AIP, Physical Properties Of Polymers Hand Book, Williston, Vt, 1996.

Semester – VII: Departmental Elective – V

Subject Code: KPE 075	Biodegrdable Polymers	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Understand various methods of biodegradation of polymers	K2
CO-2	Understand various aspects of starch based products	K2
CO-3	Acquire knowledge of bio-polyesters and their applications	K2
CO-4	Acquire knowledge of the recycling of biodegradable polymers	K2
CO-5	Acquire knowledge about various test methods and standards for biodegradable polymers	K2

UNIT I

Chemistry and Biochemistry of Polymer Degradation

Introduction, enzymes – enzyme nomenclature – enzyme specificity – physical factors affecting the activity of enzymes – enzyme mechanism, Chemical degradation initiated biodegradation, Hydrolysis of synthetic biodegradable polymers.

UNIT II

Particulate Starch Based Products

Development of Technology, Current objectives, relative starch technology, Manufacture of master batch, Conversion technology – processing precautions – moisture and temperature – rheological considerations, cyclic conversion process, physical properties of products – sample preparation – physical testing methods – test results, Quality control testing of degradation – auto oxidation measurement – biodegradation assessment – soil burial test. Introduction

UNIT III

History of Bio-polyesters

Biosynthesis, Isolation – solvent extraction - sodium hypo chloride digestion, enzymatic digestion, Properties – crystal structure and morphology.

UNIT IV

Recycling Technology for Biodegradable Plastics

Introduction, conventional recycling – economic incentive – recycling problems, degradable complicate recycling – polyethylene/starch film, reprocessing polyethylene/corn starch film scrap – learning to reprocess PE/S - Calcium oxide moisture scavenger – temperature control – accounting for pro-oxidant – handling PE/S repro – economics of in-plant recycling, Using PE/S repro – comparative study of PE/S repro on film properties, recycling other degradables.

UNIT V

Test Methods & Standards for Biodegradable Plastics

Introduction, defining biodegradability, criteria used in the evaluation of biodegradable polymers, tiered systems for evaluating biodegradability, choice of environment, choosing the most appropriate methodology, description of current test methods – screening test for ready biodegradability, tests for inherent biodegradability, tests for simulation studies, other methods for assessing biodegradability – petri-dish screen – environmental chamber method – composting methods - soil burial tests.

Reference Books:

1. G.J.L Griffin Blackie (ed.), Chemistry & Technology of biodegradable polymers Academic & Professional London 1994.
2. Yoshiharu Doi , Kazuhiko Fukuda(ed.) Biodegradable plastics & Polymers Elsevier 1994
3. Abraham J. Donb & others (ed.) Handbook of Biodegradable polymers, Harvard academic publishers Australia 1997.

Semester – VII: Departmental Elective – V

Subject Code: KPE 076	Rubber Technology	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Acquire knowledge about various aspects of Natural Rubbers	K2
CO-2	Apply the knowledge of compound design and vulcanisation	K2
CO-3	Understand various aspects of synthetic elastomers	K2
CO-4	Understand various thermoplastic elastomers	K2
CO-5	Understand various applications of Rubbers	K2

UNIT I

Natural Rubber

Tapping latex, Processing of Latex - Dry rubber production (Smoked sheet, air dried sheet, Crepe etc.) - Grading of rubbers - Modified natural rubber, Reclaimed rubber - process of reclamation – applications.

UNIT II

Compounding Design and Vulcanization

Sulphur vulcanization and non-sulphur vulcanization, vulcanization systems - accelerators, activators, promoters, antioxidants, antiozonants, processing aids, fillers and effect of fillers, Blowing agents etc.

UNIT III

Synthetic Elastomers

Manufacturing, structure, properties, compounding, curing and applications - Polyisoprene, Polybutadiene, SBR, EPDM, Butyl rubber, Neoprene, Nitrile rubber, Silicone rubber, Fluoroelastomer, Polysulphide rubber, polyurethane rubber, Acrylic rubber.

UNIT IV

Thermoplastic Elastomers

Basic structure, Manufacture, Morphology, Commercial grades and Applications – Thermoplastic styrene block copolymers, Polyester thermoplastic elastomers, polyamide thermoplastic elastomer, Polyurethane thermoplastic elastomers.

UNIT V

Rubber Product Manufacturing

Manufacturing of Belting, Hoses, Footwear, Rubber metal bonded items, sports goods, cellular rubber, tyres etc.

Text Books:-

1. C.M.Blow and Hepburn, - Rubber Technology and Manufacture, 2nd edition, 1982.
2. Hoffman, Rubber Technology Handbook - Hanser Pub. Munich – 1996

References :-

1. Anil .K. Bhowmic, Howard L. Stephens (Edt), Handbook of Elastomers – New Developments & Technology, Marcel Decker Inc. New York 1988.
2. Maurice Morton, Rubber Technology,1998.

Semester – VII: Departmental Elective – V

Subject Code: KPE 077	Polymeric Nanomaterials	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Understand basics of Polymeric nanoparticles	K2
CO-2	Understand various types of nanocomposites of polymers and inorganic particles	K2
CO-3	Understand various aspects of carbon nanomaterials	K2
CO-4	Acquire knowledge of various applications of nanomaterials	K2
CO-5	Acquire knowledge about various methods of characterisation of nanomaterials	K2

UNIT I

Polymeric nanoparticles

Introduction, synthetic routes for polymeric nanoparticles, super critical fluid based particle production, droplet and aerosol techniques, gas atomization approaches dendrimers, hyper branched polymers or star polymers, molecular imprint polymers, applications of polymeric particles. Nano clays, nano oxides, nanowires, nanotubes and nanofibres, polymer nanofilm, nanostructured polymers with special architectures.

UNIT II

Nanocomposites of polymers and inorganic particles

Introduction, preparation of nanocomposites, characterization and properties of Nanocomposites, structure of clay and its modification with surfactants, preparative methods and structure of polymer/clay nanocomposites, types of polymers used for polymer/clay nanocomosites preparation, material properties of polymer/clay nanocomposites, processing operations of nanocomposites.

UNIT III

Carbon nanomaterials

Structural aspects, preparation of nano tubes: carbon arc process, catalytic assisted pyrolysis, laser technique, electro chemical method, purification of carbon nano tube, properties of nano tubes, Single walled nano tubes, multi walled nano tubes, application of nanotubes, CNT- polymer-matrix composites – processing of polymer nano composites - properties of polymer nanocomposites.

UNIT IV

Applications of Nanoparticles

Introduction, features of polymeric materials, preparation and characterization of nanoparticles, recent developments in nanoparticles technology, nanoparticles for some specific applications.

UNIT V

Characterisation of Nanomaterials

Introduction, Morphological characterisation, Physico-chemical characterisation, Atomic Force Microscopy, Small angle X-ray scattering, Raman Spectroscopy, Dynamic light scattering, Brunauer-Emmett-Teller-Method, Nanoparticle Tracking Analysis, Ultra Centrifugation, Electrophoresis.

Reference Books:-

1. Y.C. Ke, P. Stroeve and F.S. Wang**,** Polymer layered silicate and silica nano composites, Elsevier, 2005.
2. B. K. G. Theng. Formation and properties of clay-polymer complexes. Elsevier, Amsterdam, 1979.
3. B.K.G. Theng. Chemistry of clay-organic reactions. Wiley, New York, 1974.
4. V.Chirala, G.Marginean, W.Brandl and T.Iclanzan, Vapour grown carbon nanofibres-polypropylene composites and their properties in Carbon nanotubes edited by V.N. Popov and P.Lambin, Springer (2006), Netherlands.
5. Recent Advances in Polymer Nanocomposites; Editors: S. Thomas, G.E. Zaikov and S.V. Valsaraj, CRC Press, 2009

Semester – VII: Departmental Elective – V

Subject Code: KPE 078	Thermoplastic Elastomers	L T P : 3 0 0	Credits: 3

Course Outcomes: The students will be able to		Blooms Taxonomy
CO-1	Understand various classifications of thermoplastic elastomers	K2
CO-2	Understand different thermoplastic elastomers from conventional polymers	K2
CO-3	Understand various aspects of Polyurethane elastomers	K2
CO-4	Acquire knowledge of various aspects of Polyamide and Polyether based Elastomers	K2
CO-5	Acquire knowledge of various Thermoplastic elastomer from Blends	K2

UNIT I

Classification of Thermoplastic Elastomers

Introduction to Thermoplastic Elastomers (TPE) Polyolefin based thermoplastic elastomers – Block copolymer, Random Block polymers, Graft copolymers, Polyolefin blend TPE's, preparation, properties, processing and applications.

UNIT II

Thermoplastic Elastomers from Conventional Polymers

Polyvinylchloride based Thermoplastic Elastomers – PVC/Nitrile Rubber blends, PVC/Polyurethane blends. Styrenic Thermoplastic Elastomers – Manufacture, Properties and applications.

UNIT III

Polyurethane Elastomers

Thermoplastic Polyurethane Elastomer – Raw materials, Synthesis, Properties, Processing, Blends and their applications.

UNIT IV

Polyamide and Polyether based Elastomers

Polyamides based Thermoplastic Elastomers – Polyamide thermoplastic elastomers, Preparation, properties, and applications. Thermoplastic Polyether ester elastomers – Synthesis, Properties and applications.

UNIT V

Thermo Plastic Elastomers from Blends

Introduction - Preparation of Elastomer – Plastic blends by dynamic vulcanization, properties and applications. Ionomeric Thermoplastic Elastomers: Synthesis, Properties and applications of ionomeric elastomers.

References :-

1. Anil K. Bhowmick, Howard L. Stephens, Hand Book of Elastomers, New Developments and Technology, Marcel Dekker, Inc., New York, 1988.
2. Benjamin M. Walker, Hand Book of Thermoplastic Elastomers, Van Mostrand Reinhold Company, New York, 1979
3. G.Holden, N.R. Legge, R. Quirk, H.E. Schrolder, Thermoplastic Elastomers – 2nd Edition, Hanser Publishers, Munich, 1996.
4. S.K. De, Anil K. Bhowmick, Thermoplastic Elastomers from Rubber – Plastic Blends, Ellis Horwood, N